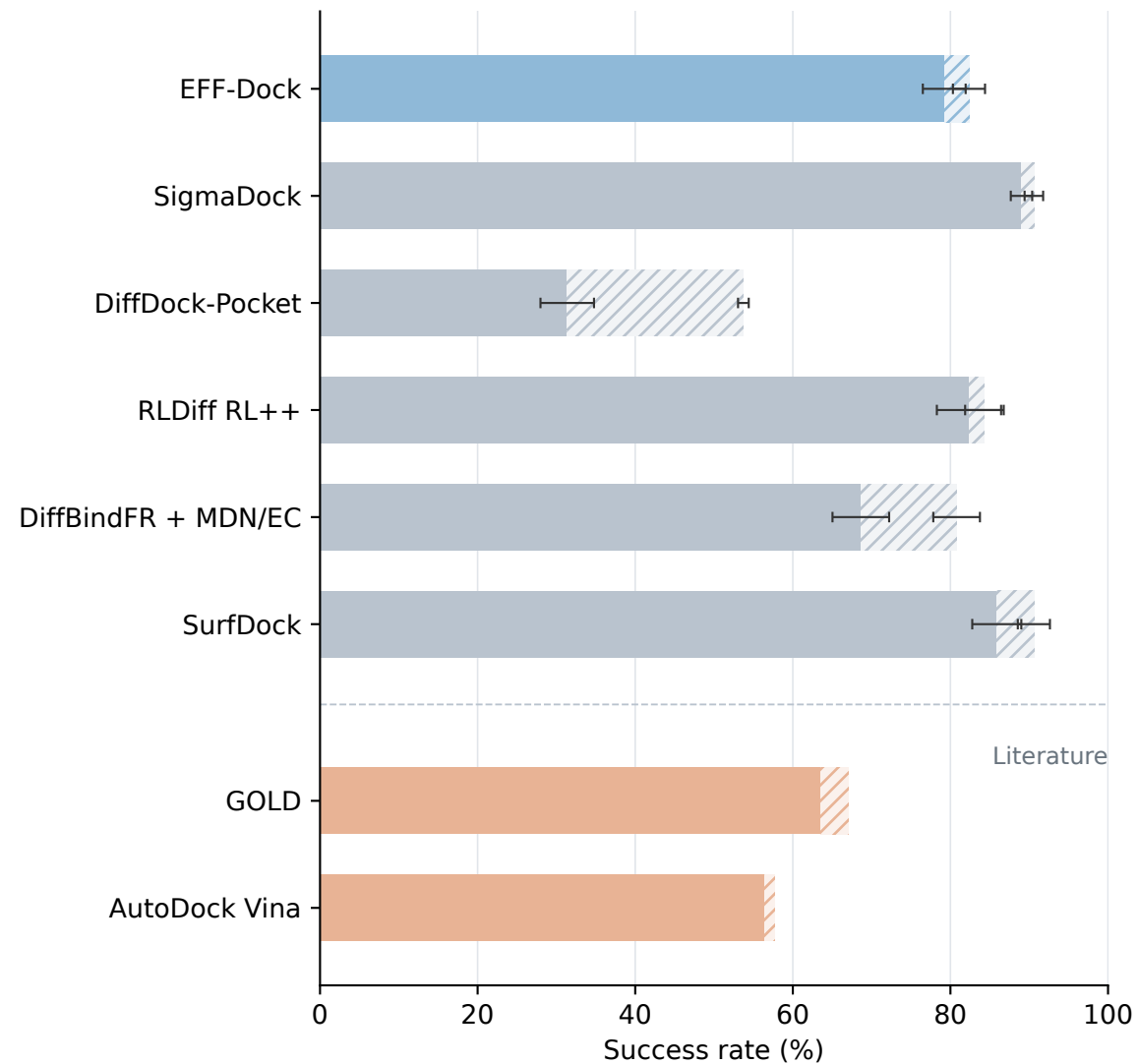
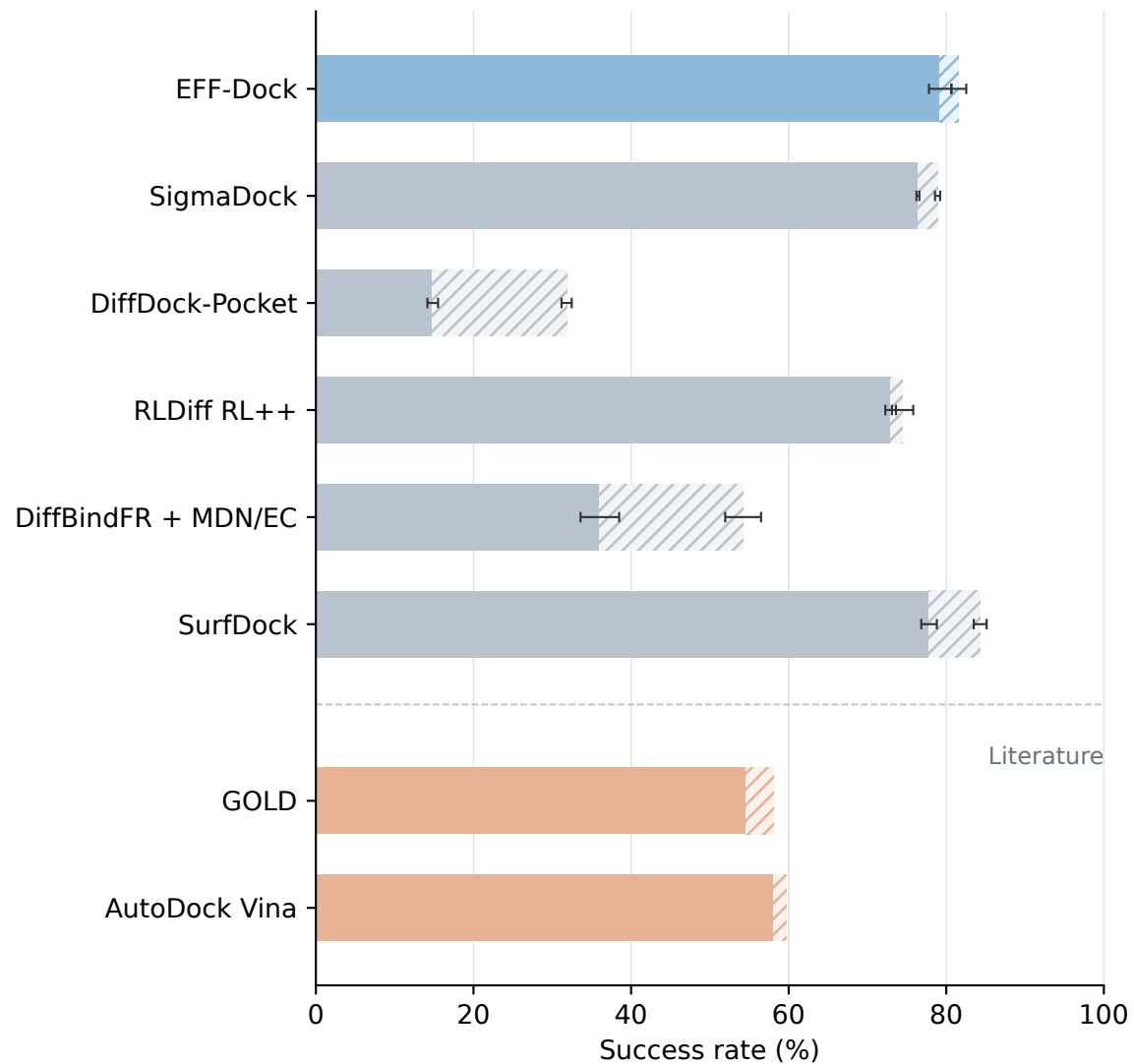
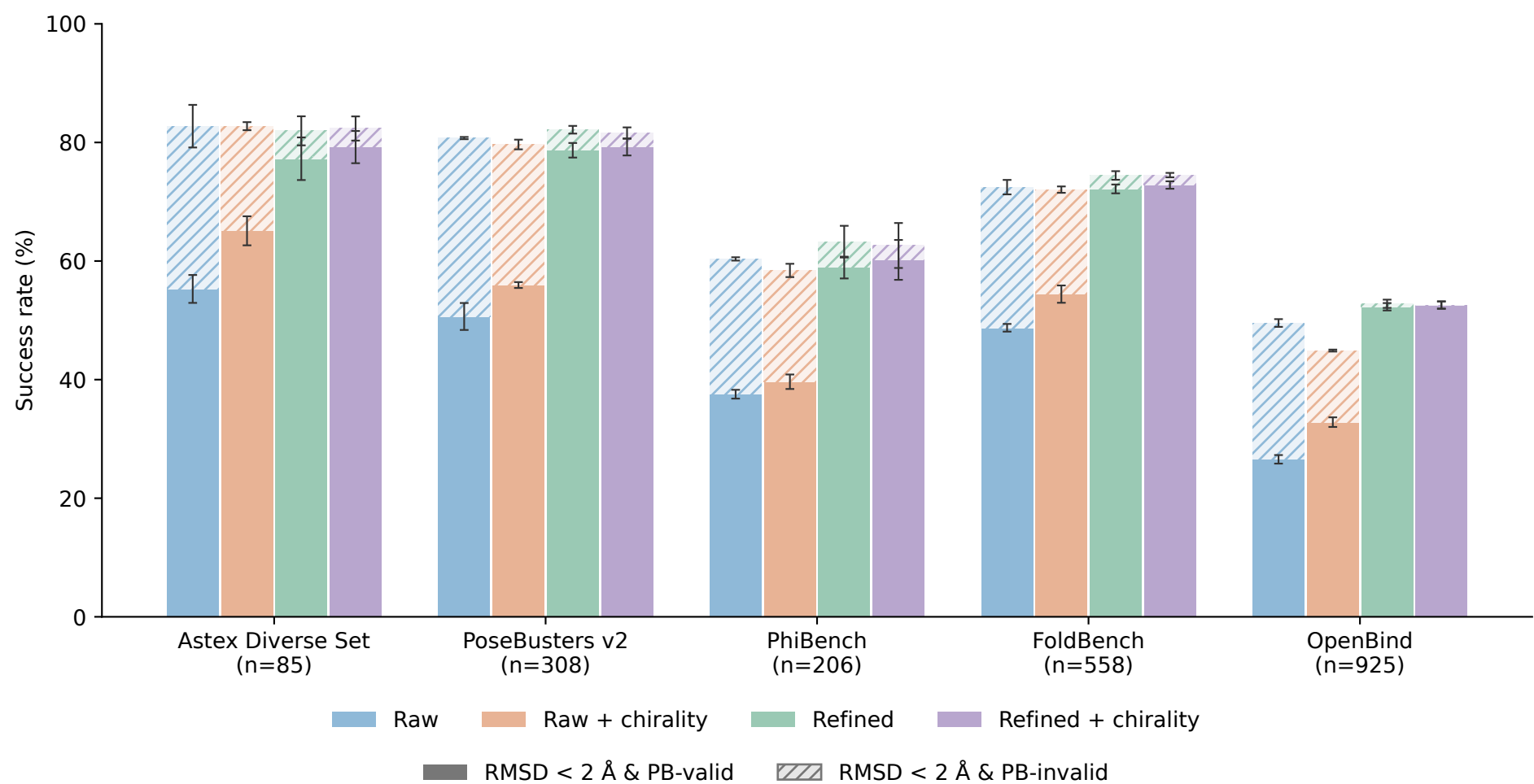
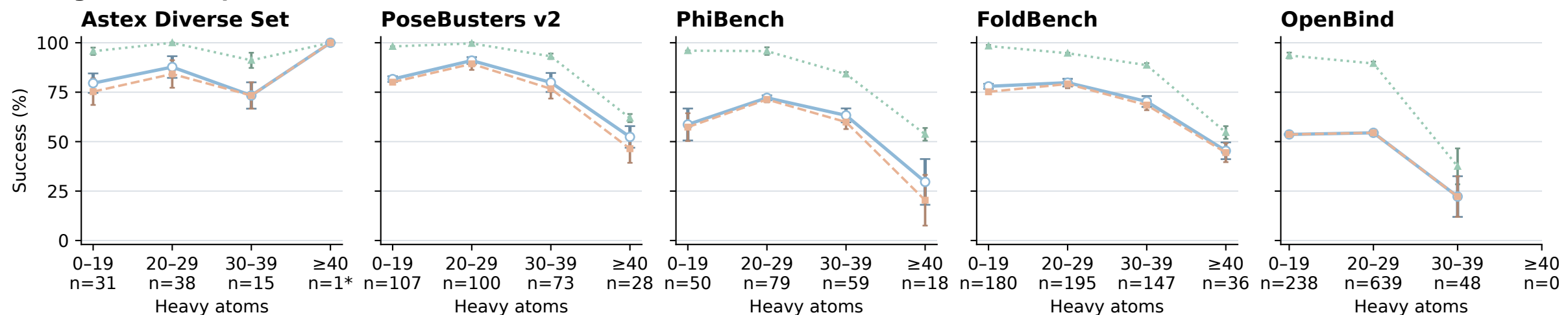


A Astex Diverse Set**B PoseBusters v2**

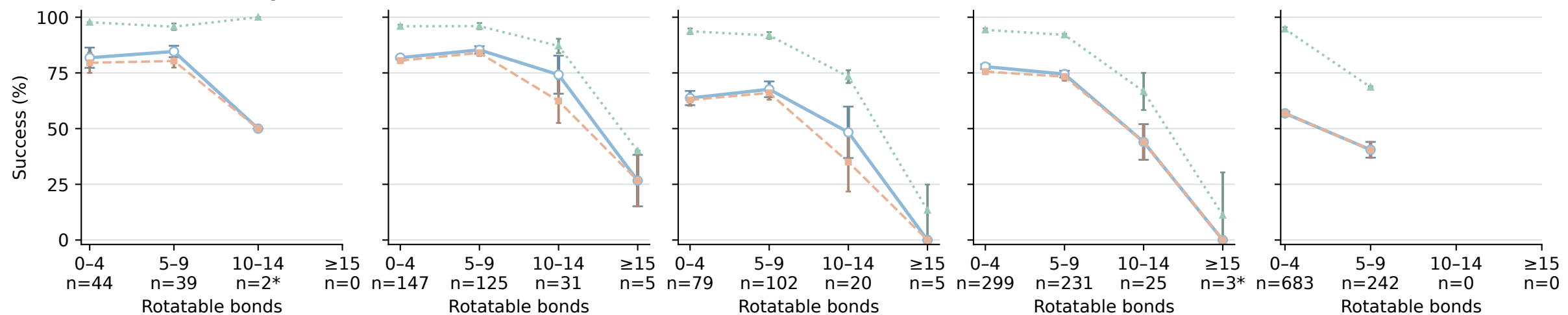
■ RMSD < 2 Å & PB-valid ▨ RMSD < 2 Å & PB-invalid



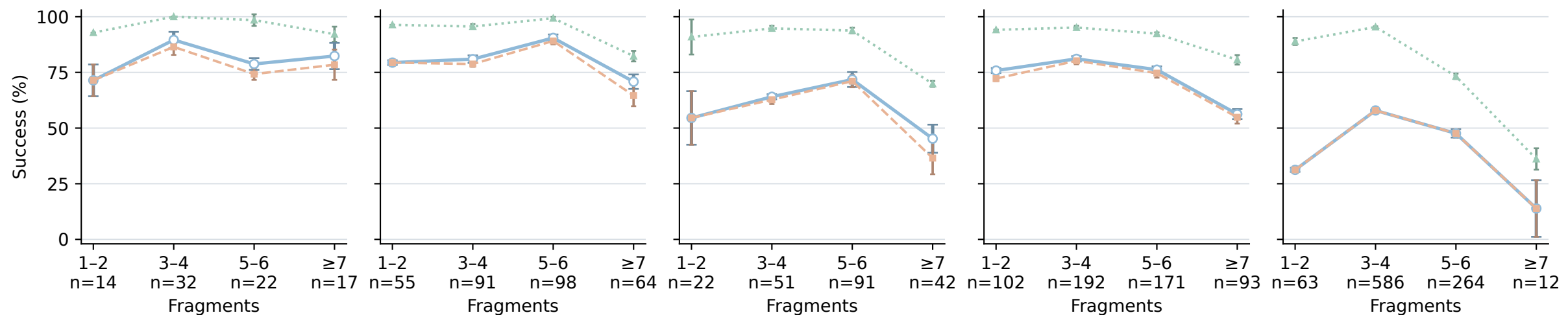
A Ligand size dependence



B Rotatable bond dependence

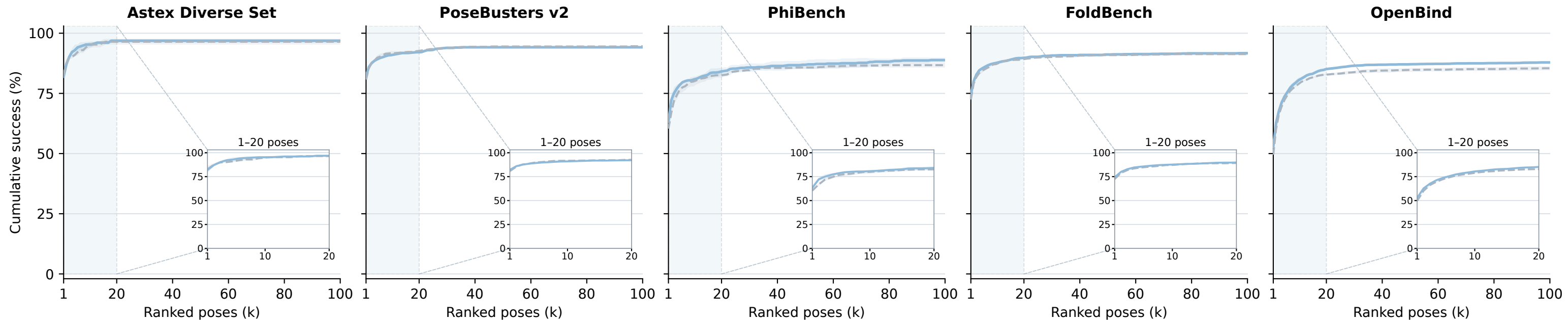


C Fragment count dependence

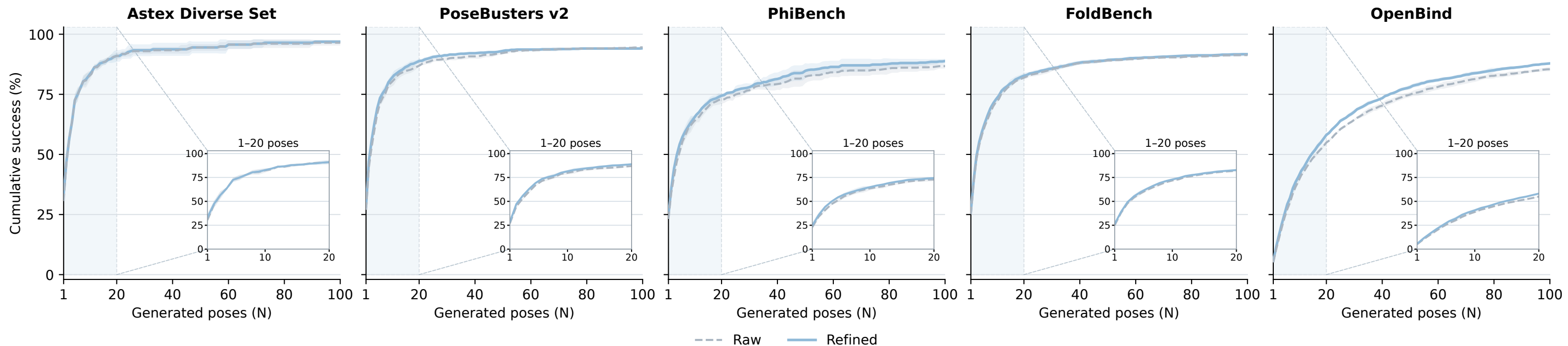


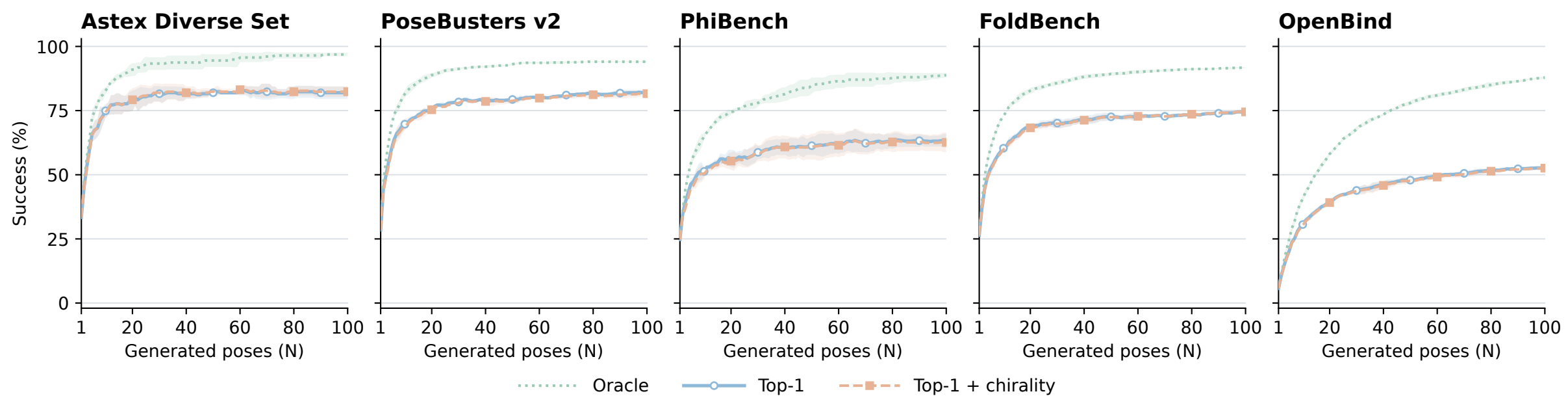
Oracle Top-1 Top-1 + PB-valid

A Confidence-ranked poses

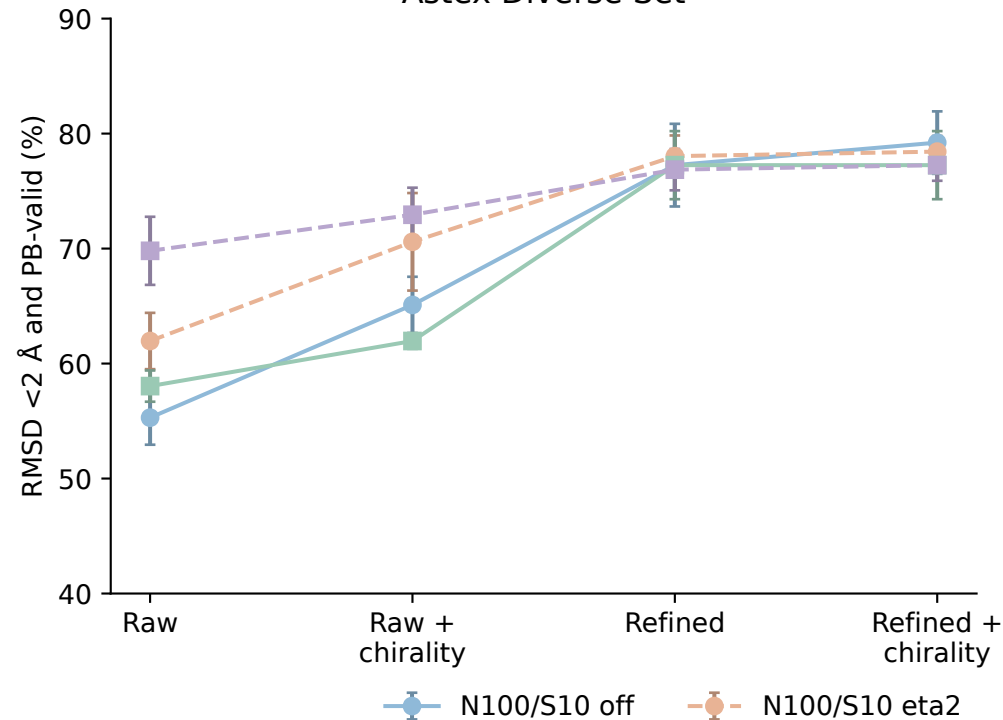


B Generation-order poses

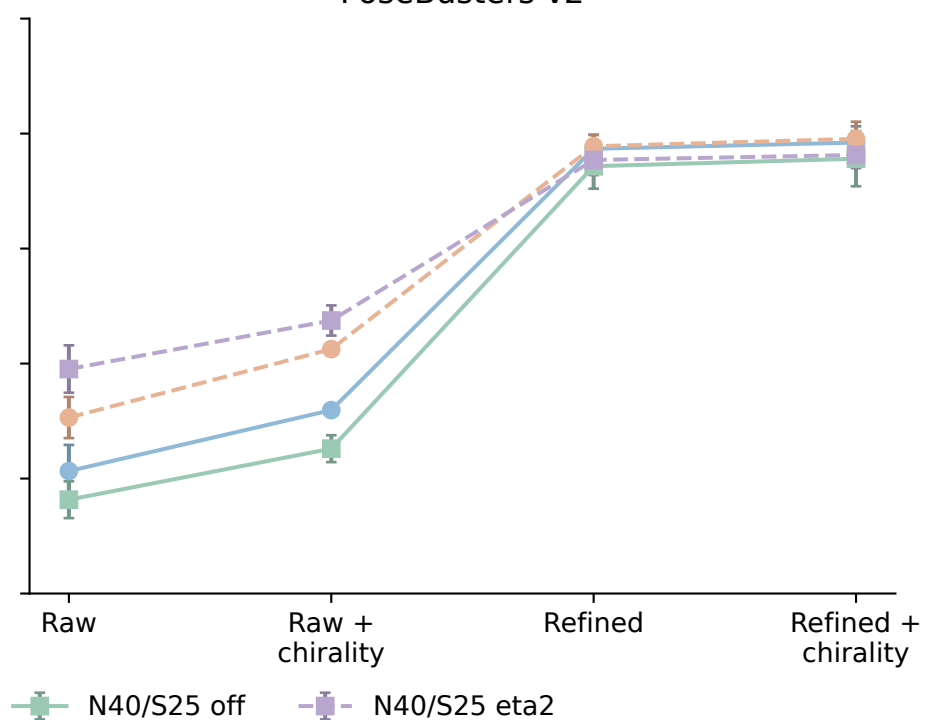


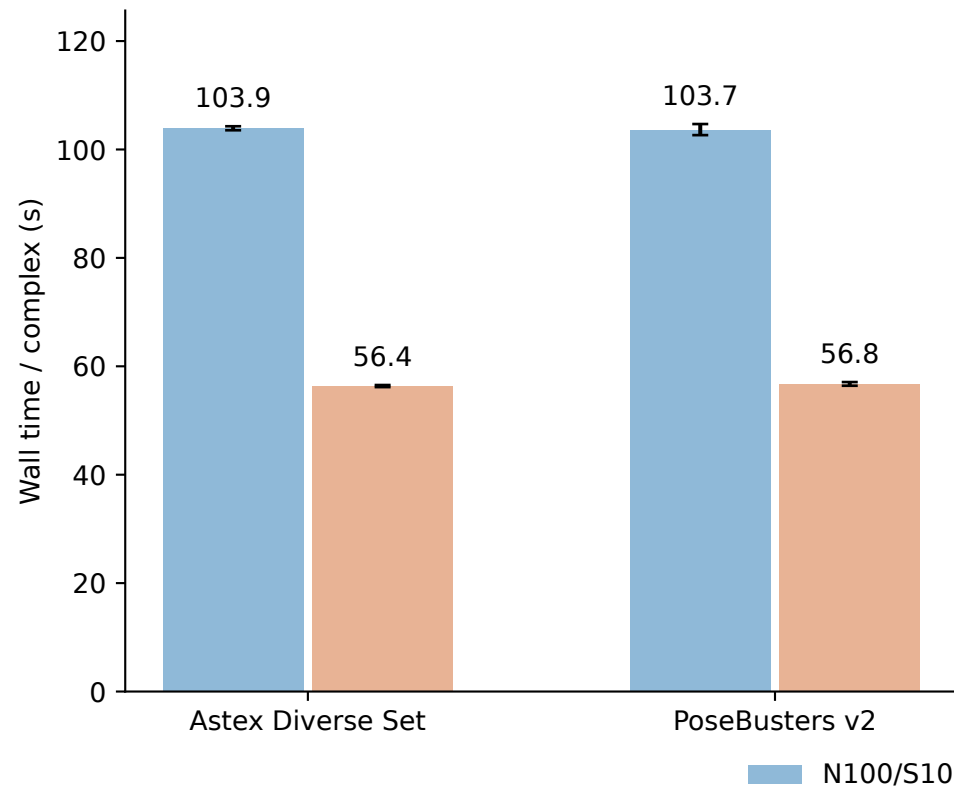
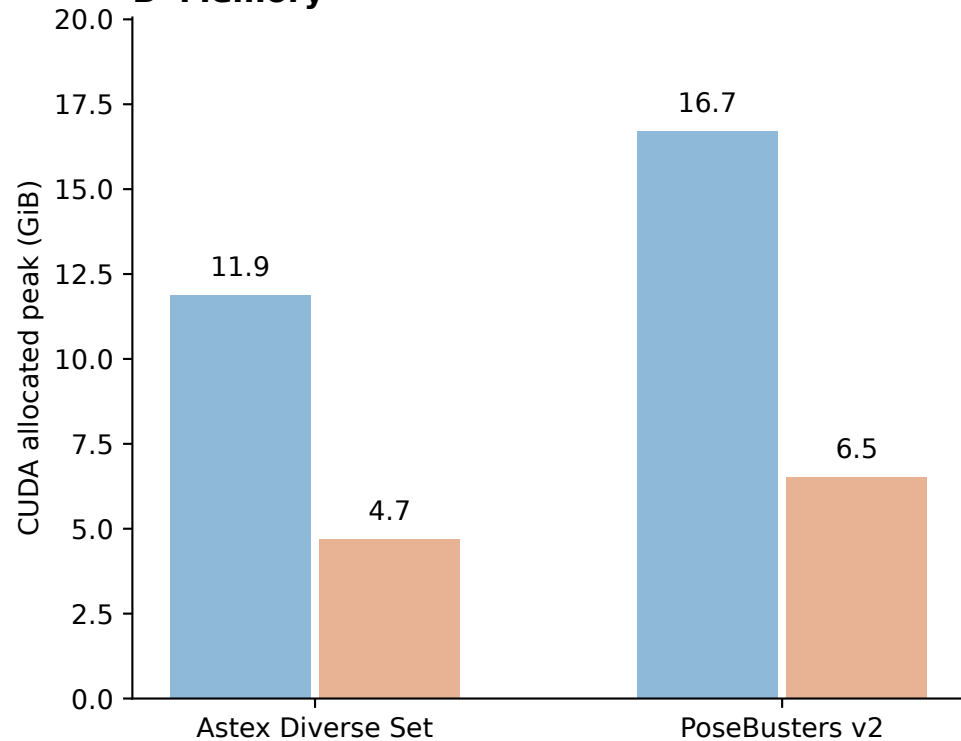


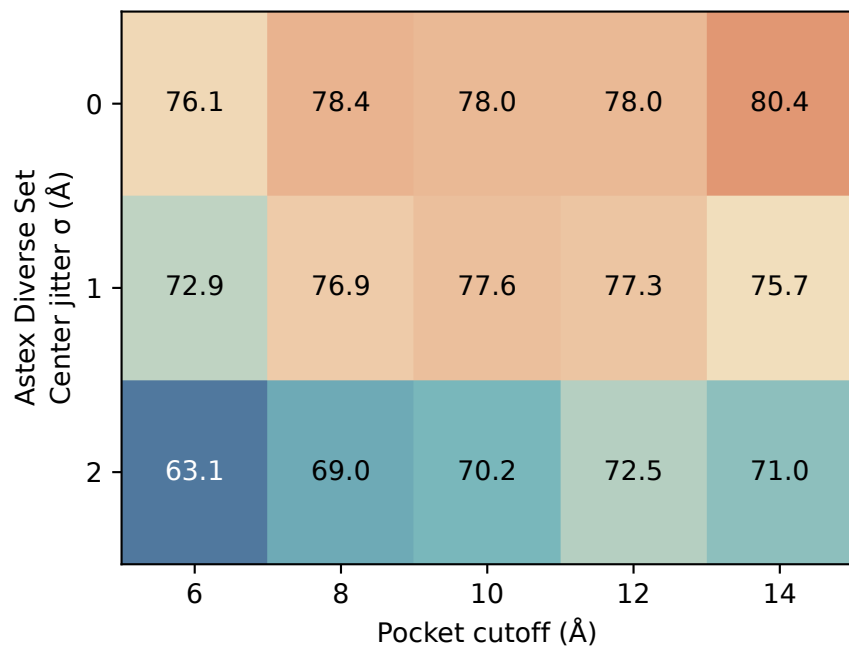
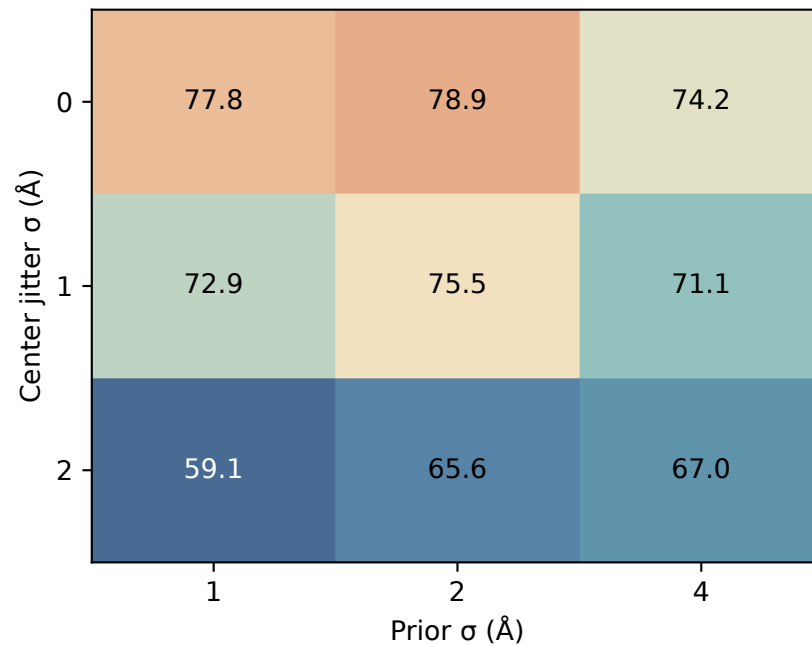
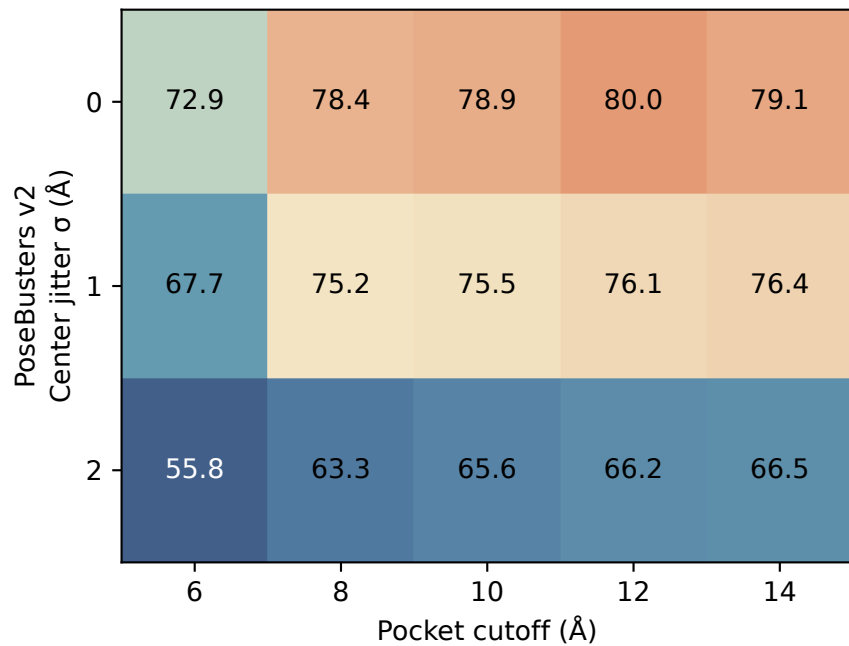
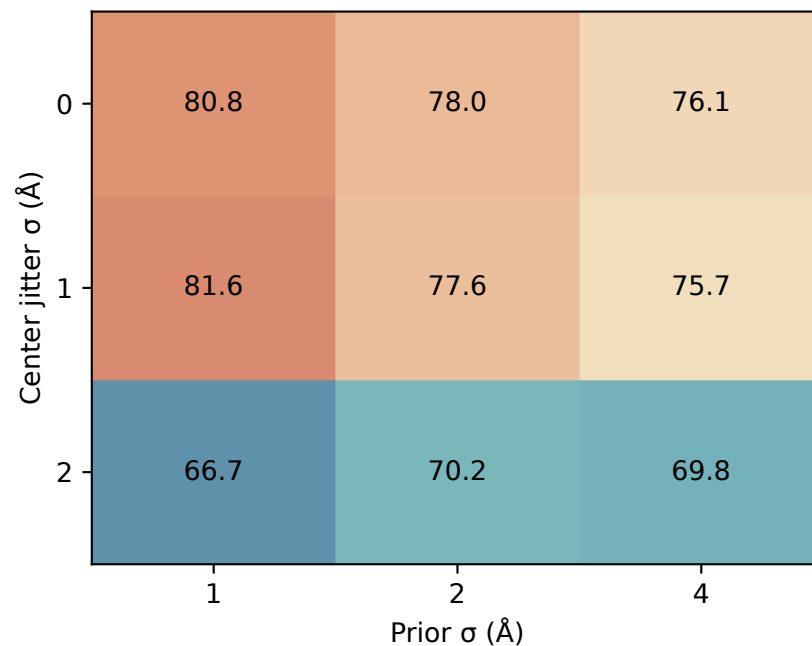
Astex Diverse Set



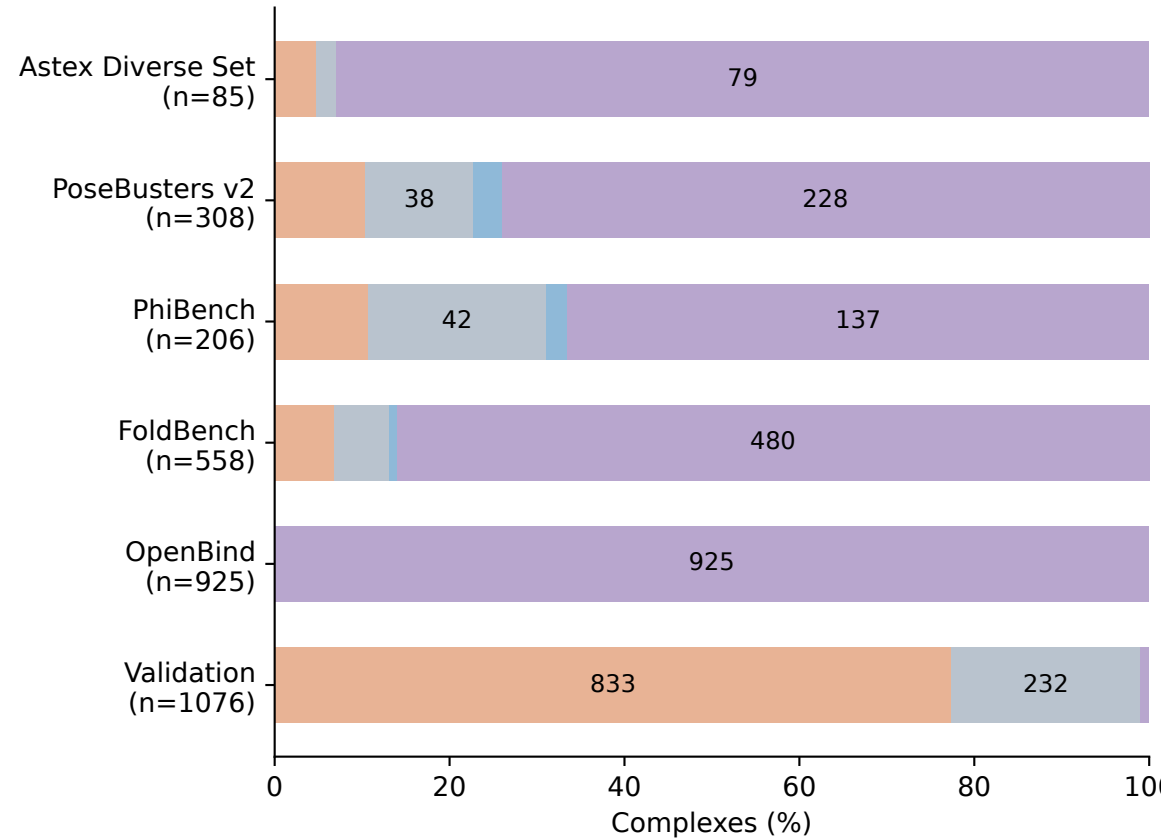
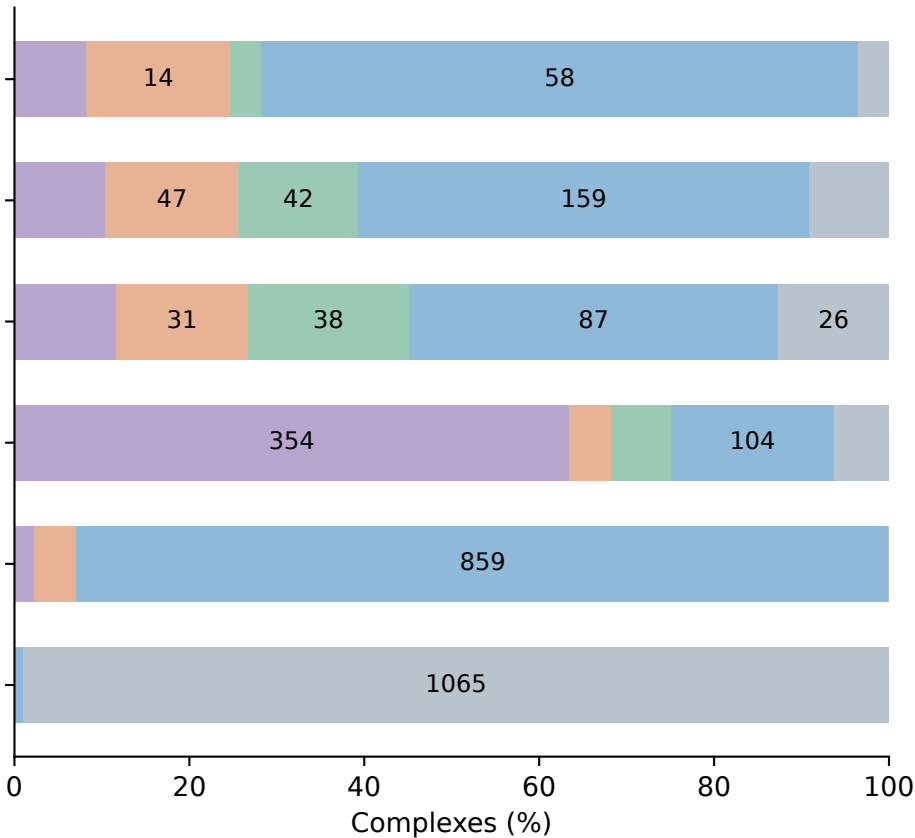
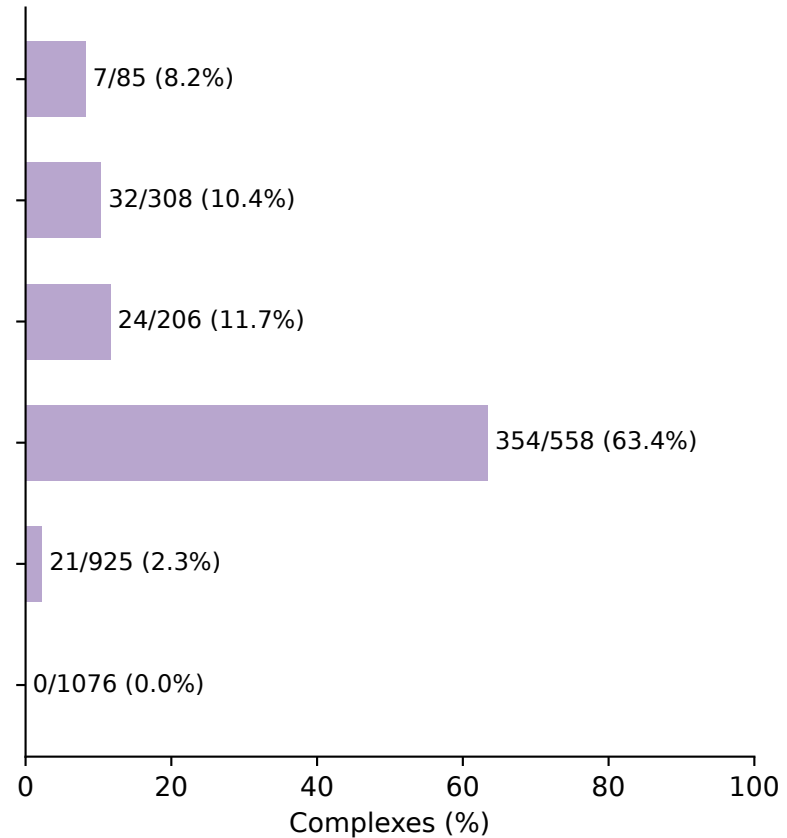
PoseBusters v2



A Runtime**B Memory**

A Pocket size**B Prior scale**

RMSD < 2 Å & PB-valid (%)

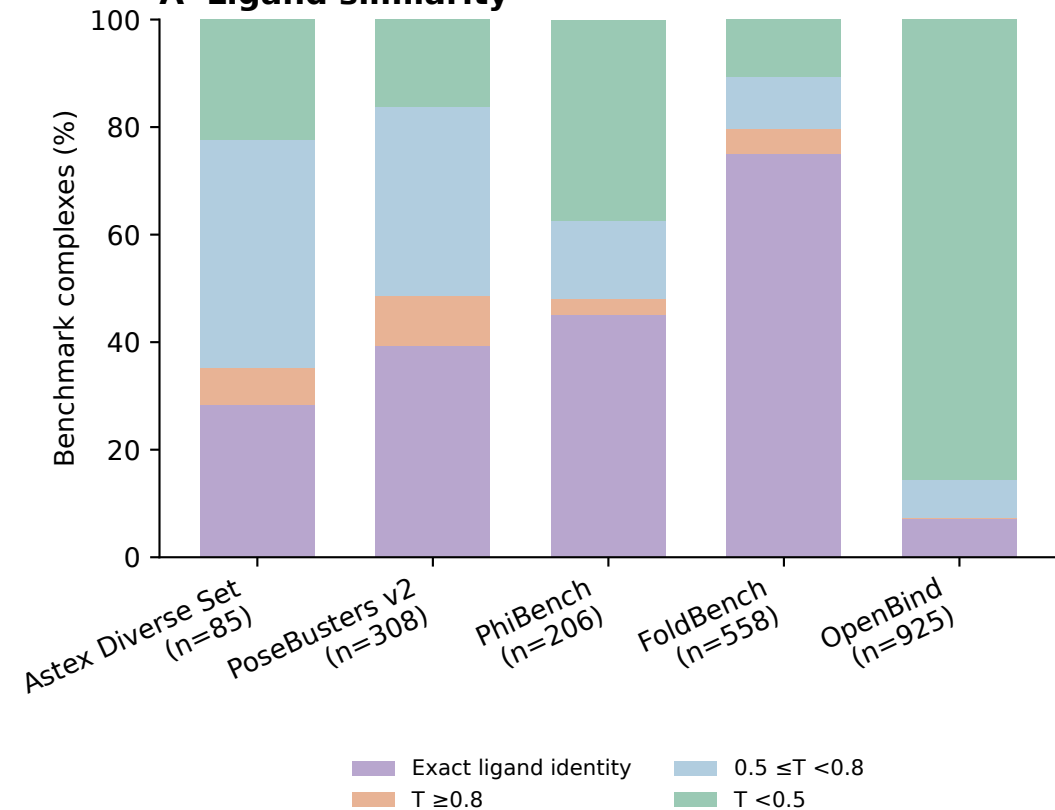
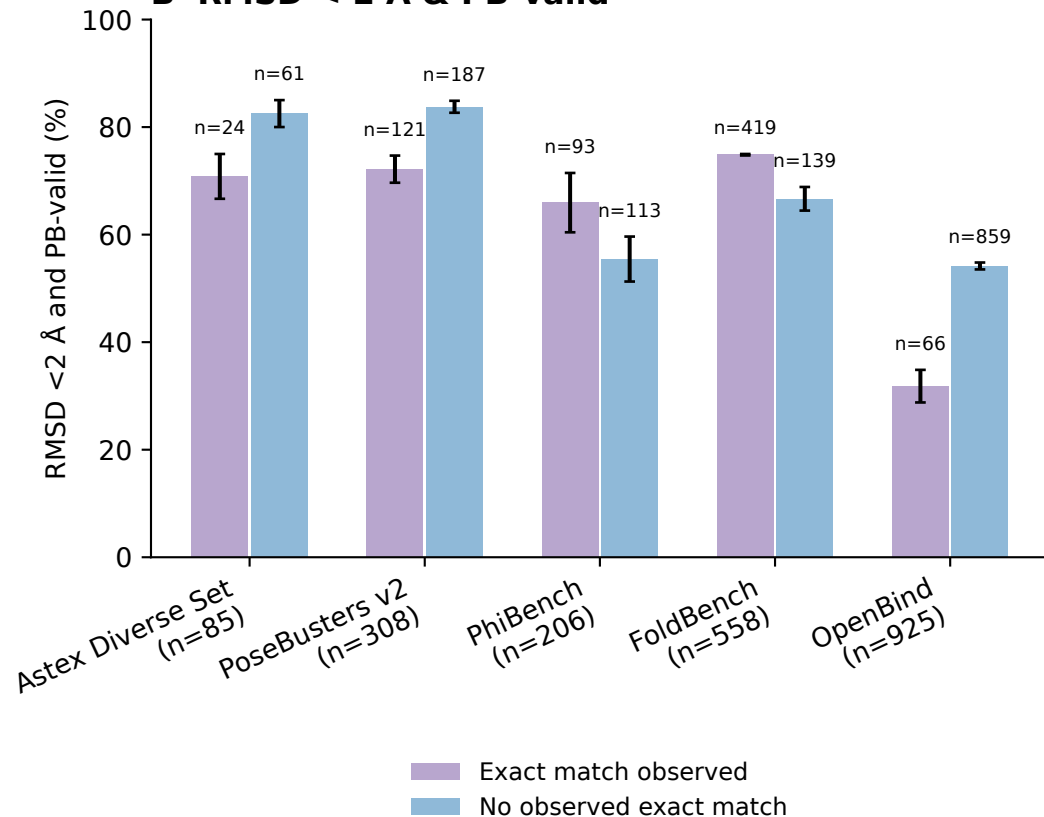
A Sequence identity**B Ligand-protein overlap****C Same training sample**

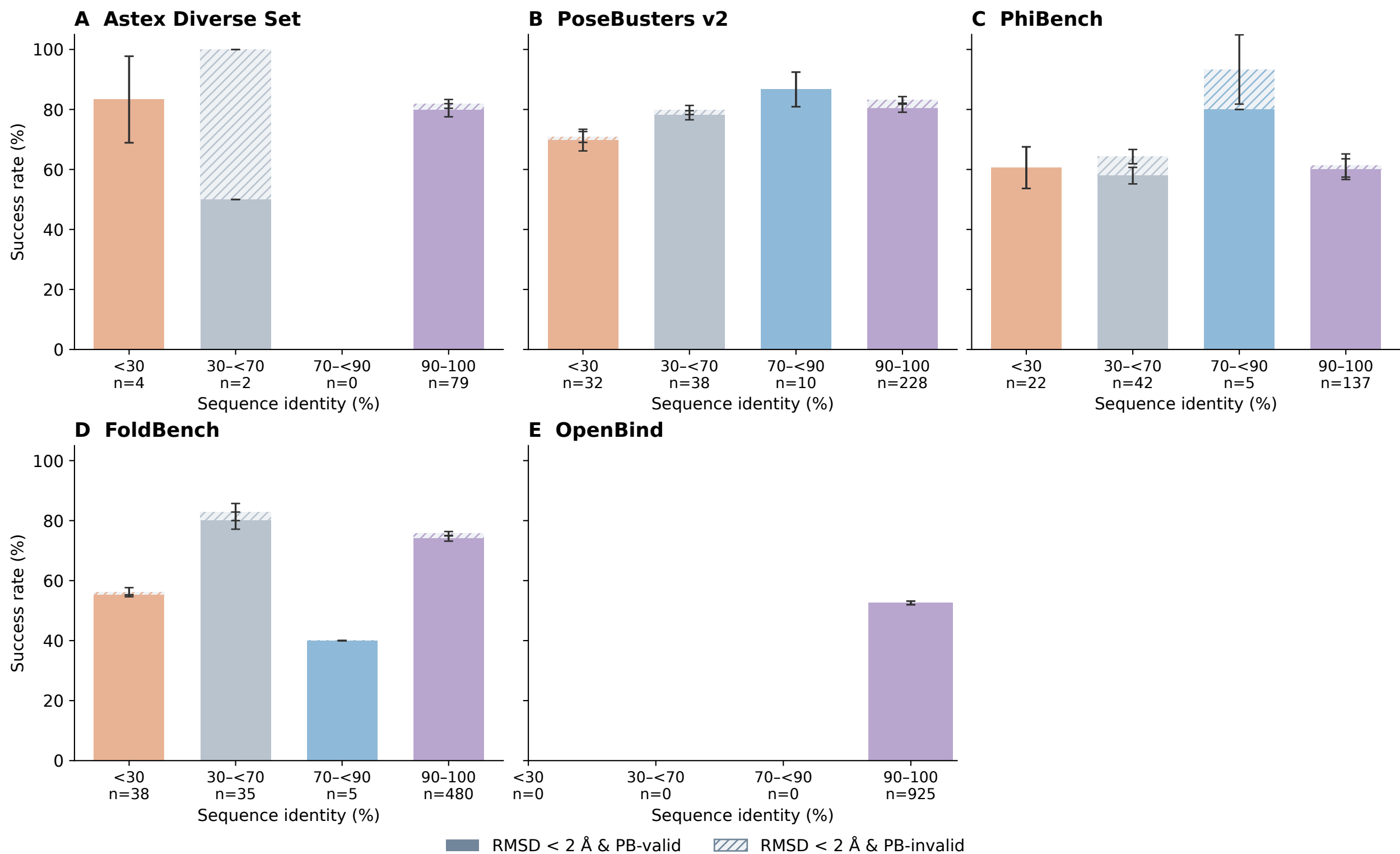
Sequence identity (%)

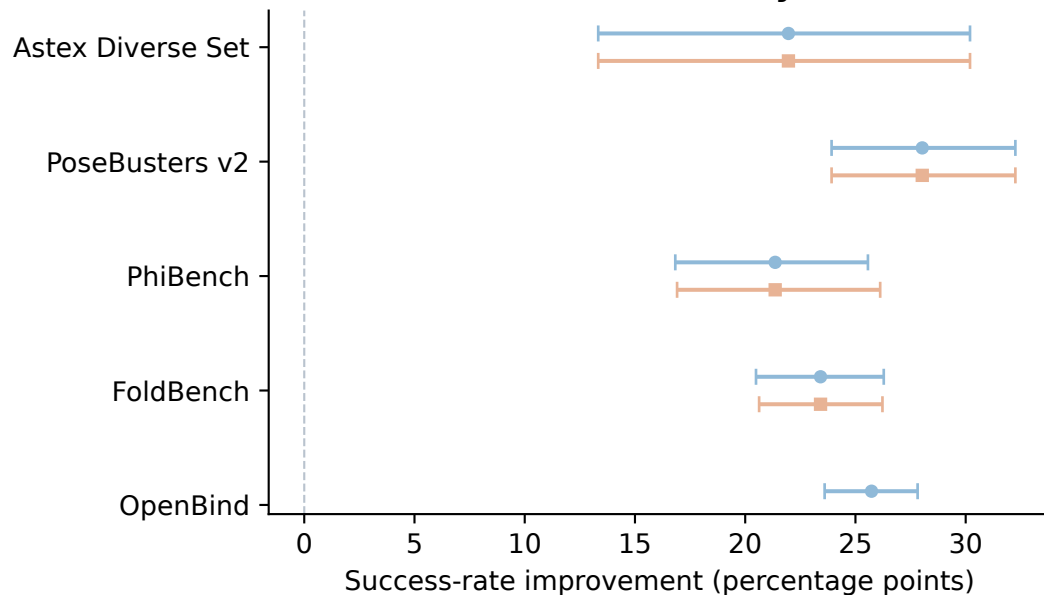
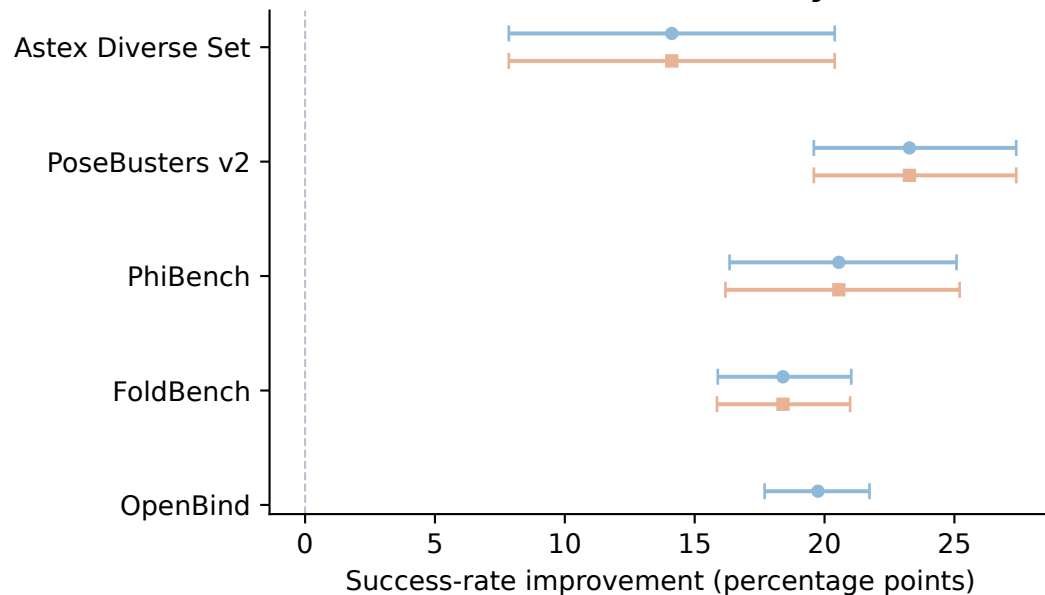
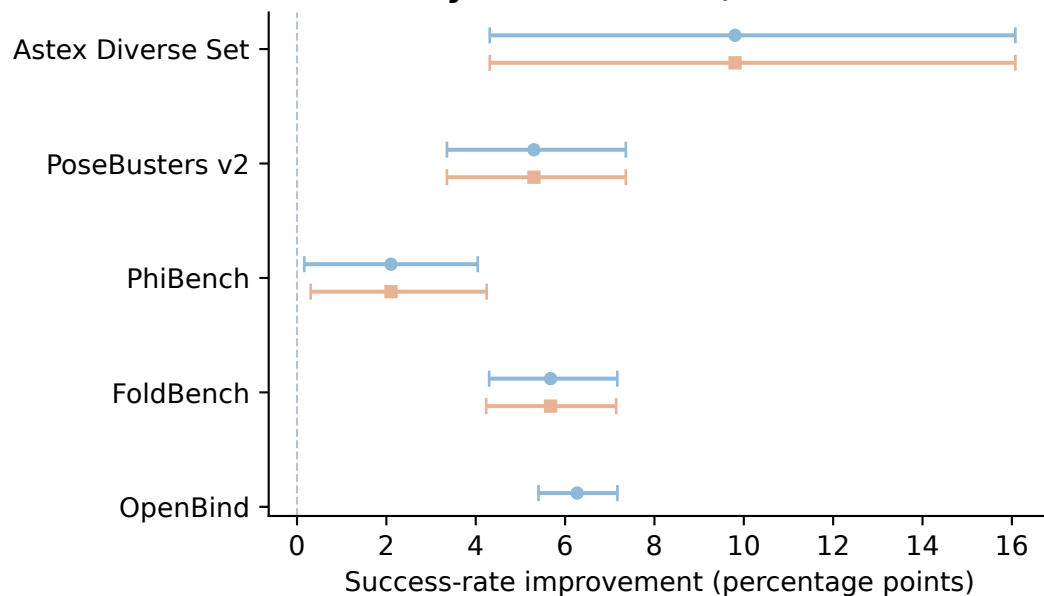
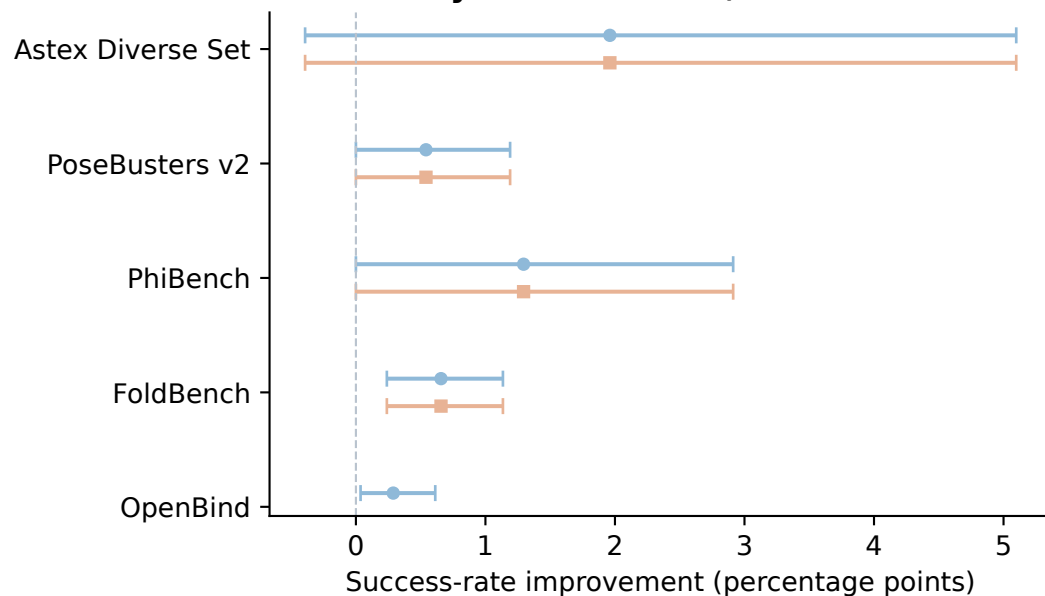
- <30%
- 30-<70%
- 70-<90%
- 90-100%

Ligand-protein overlap

- Same-sample overlap
- Separate matches
- Ligand only
- Sequence only
- Neither observed

A Ligand similarity**B RMSD < 2 Å & PB-valid**

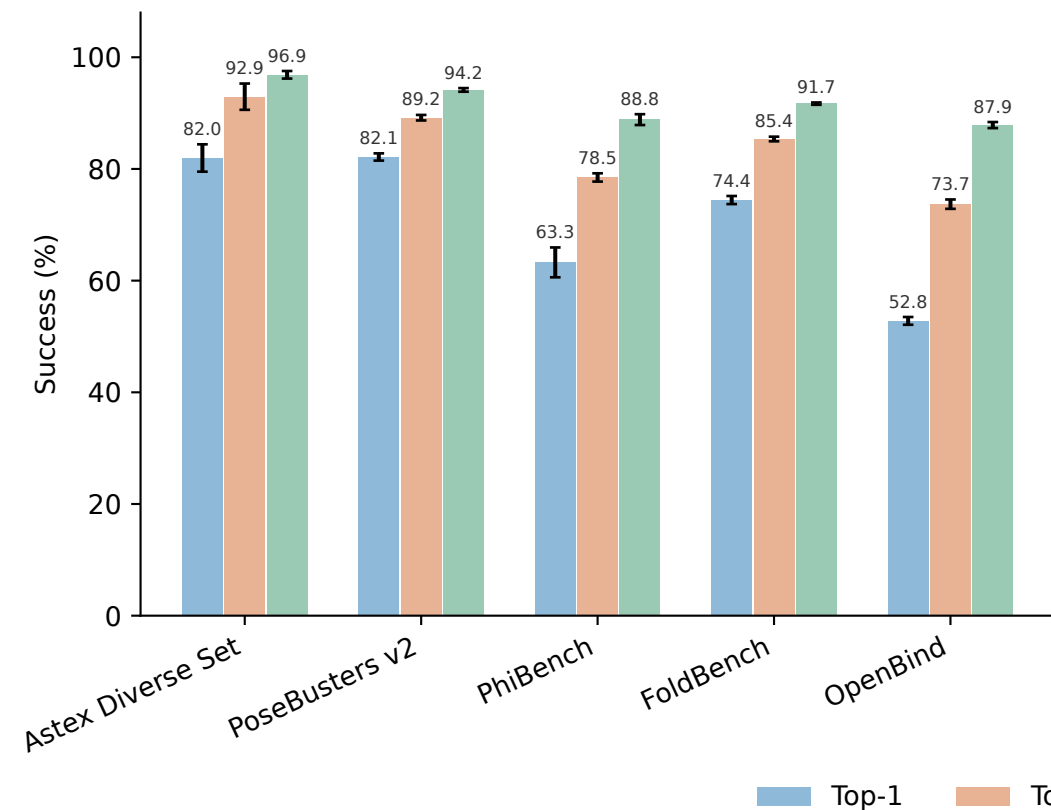
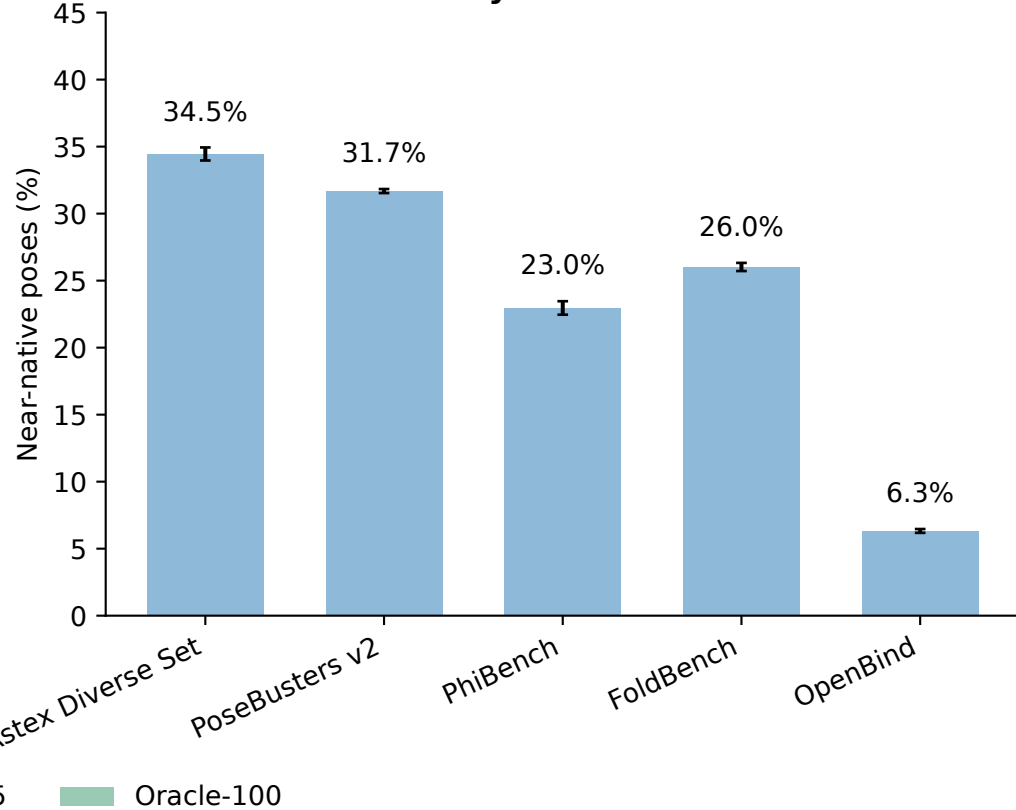


A Refined – raw, no chirality filter**B Refined – raw, with chirality filter****C Chirality filter off → on, raw****D Chirality filter off → on, refined**

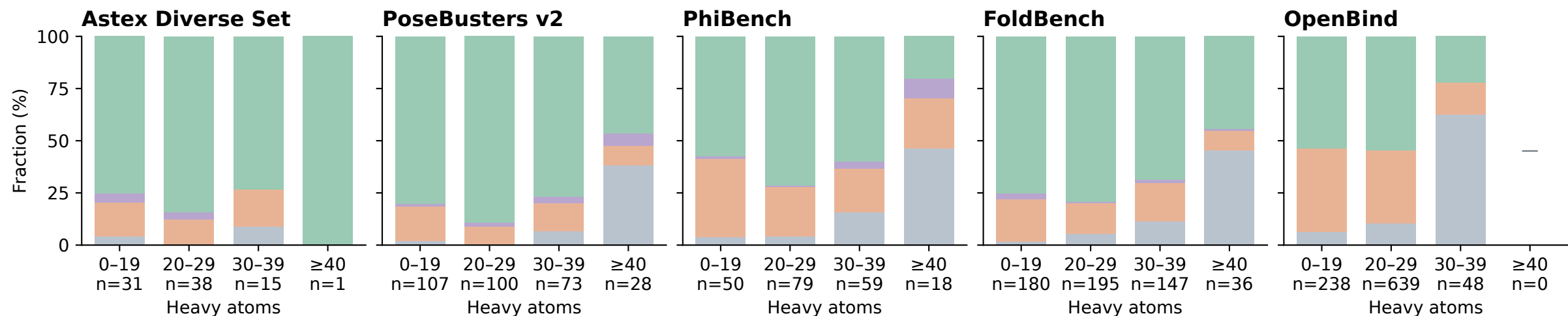
Complex bootstrap



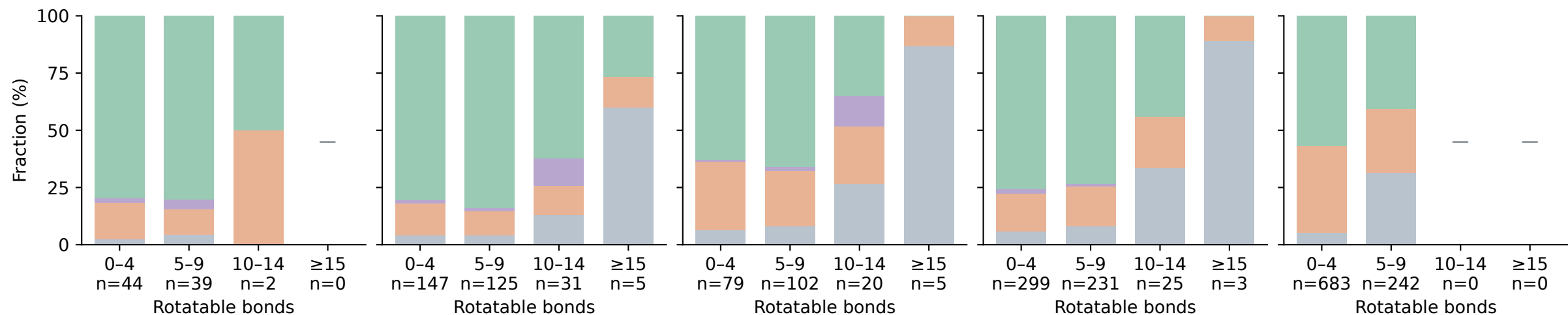
PDB-cluster bootstrap

A Selection**B Near-native density**

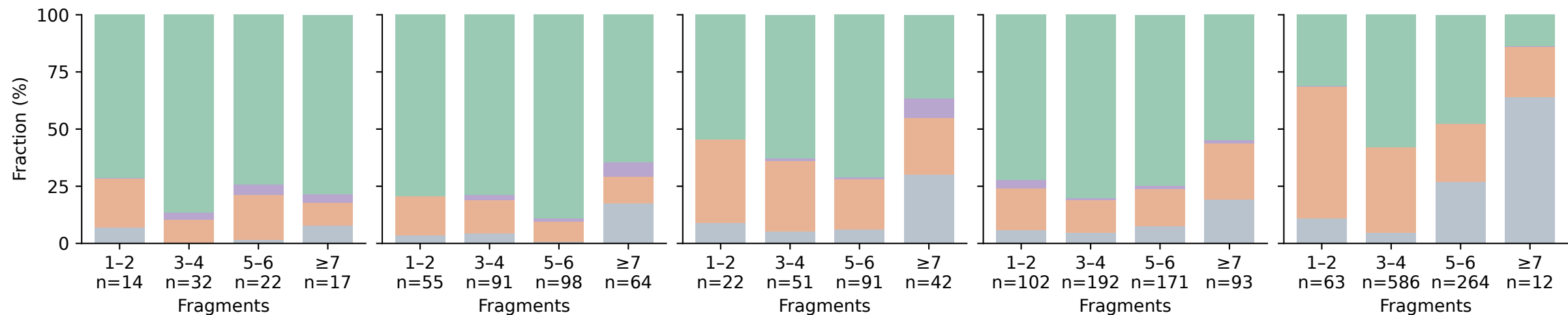
A Ligand size dependence



B Rotatable bond dependence



C Fragment count dependence



Legend: Near-native pose generation failure (blue), Selection failure (orange), PB failure (purple), Success (green)